

Modeling Spatial Dependence: A Simulation-Based Comparison of Parametric and Semi-Parametric Approaches

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March 5, 2026

Presentation Overview

- 1 Introduction
- 2 Production Function
- 3 Production Function Cont.
- 4 Parametric Spatial Models
- 5 The Spatial Weights Matrix

Why Spatial Dependence Matters

- Regional data often exhibit spatial dependence.
- Example: Puerto Rico municipalities.
- Productivity in one region may depend on neighboring regions.
- Violates classical i.i.d. assumption.

Production Function

$$Y = AK^{\alpha}L^{\beta}$$

$$\log(Y) = \log(A) + \alpha \log(K) + \beta \log(L) + \epsilon; \epsilon \sim N(0, \sigma)$$

- Y is the total output (real GDP) produced.
- A is total factor productivity (TFP), capturing the efficiency with which inputs are used.
- K is the quantity of physical capital used in production.
- L is the quantity of labor employed.

Production Function

$$Y = AK^{\alpha}L^{\beta}$$

$$\log(Y) = \log(A) + \alpha \log(K) + \beta \log(L) + \epsilon; \epsilon \sim N(0, \sigma)$$

- α is the output elasticity of capital
- β is the output elasticity of labor
- i municipalities id
- t time period

Spatial Autoregressive (SAR) Model

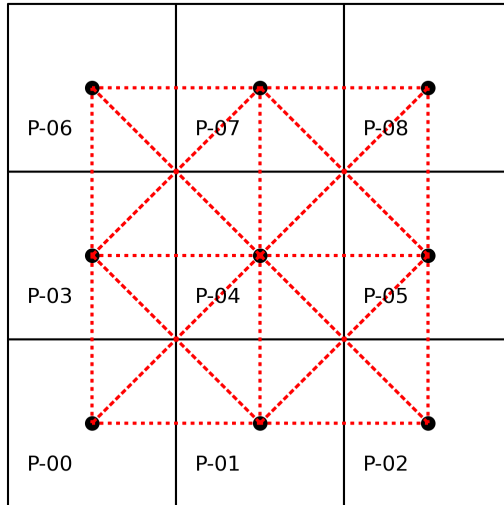
$$y = \rho Wy + X\beta + \varepsilon$$

- ρ measures spatial dependence.
- W defines neighborhood structure.
- Sensitive to correct specification of W .

Defining W : The Core Challenge

- W is assumed known in theory.
- In practice, it must be chosen.
- No universally accepted selection rule.
- Misspecification leads to biased estimates.

Example of the Weights Matrix (Queen)



Example of the Weights Matrix (KNN)

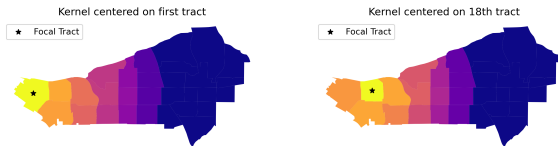


Figure: K-Nearest Neighbors (KNN) Model

Tensor Product Spatial Smoother

$$\log(Y_{it}) = \alpha + \beta_1 \log(K_{it}) + \beta_2 \log(L_{it}) + f(\text{lat}_i, \text{lon}_i) + \varepsilon_{it}$$

Key Idea

No need to specify W . Spatial effects captured flexibly using splines.

Simulation Results (1000 Runs)

Variable	Rook	Queen	Tensor	OLS
Intercept	3973.3	0.40	8252.6	7177.3
X_1	0.74	0.0032	0.21	1.22
X_2	0.90	0.0032	0.26	1.57
X_3	1.04	0.0031	0.27	1.92

Finding: When W is misspecified, the tensor approach is substantially more robust.

Main Conclusions

- Parametric models are highly sensitive to W .
- Misspecification leads to large estimation errors.
- Semi-parametric tensor approach provides robust spatial control.
- Useful in contexts like Puerto Rico where W is uncertain.